

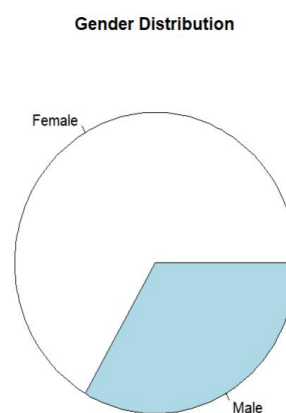
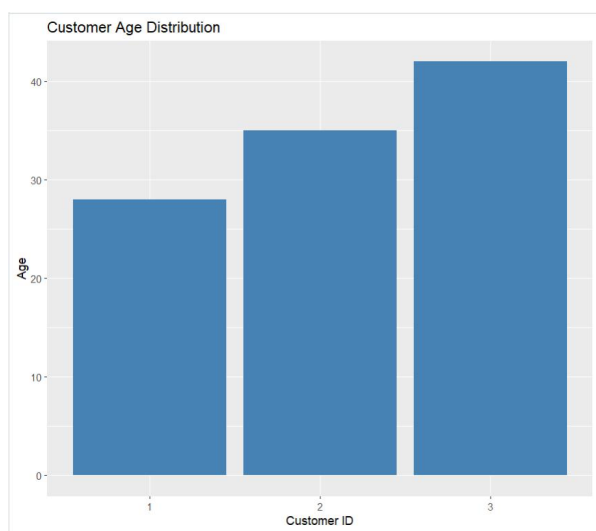
SET 16

Customer Demographics Analysis

Dataset (CSV)

Procedure

1. Create the customer demographic dataset containing age, gender, and income.
2. Import the dataset into R Studio.
3. Load the required libraries for visualization and data presentation.
4. Generate a bar chart to visualize customer age distribution.
5. Create a pie chart showing gender distribution.
6. Build an interactive table displaying demographic details.
7. Combine the visualizations into a dashboard structure.
8. Study demographic patterns that can support marketing decisions.



R Code

```
library(ggplot2)
```

```
library(plotly)
```

```
library(DT)
```

```
customer <- data.frame(  
  CustomerID=c(1,2,3),  
  Age=c(28,35,42),  
  Gender=c("Female","Male","Female"),  
  Income=c(50000,60000,75000)  
)
```

```
# Q1 Bar Chart
```

```
ggplot(customer,  
  aes(factor(CustomerID),Age))+  
  geom_bar(stat="identity",fill="steelblue")+  
  labs(title="Customer Age Distribution",  
    x="Customer ID",  
    y="Age")
```

```
# Q2 Pie Chart
```

```
gender_count <- table(customer$Gender)
```

```
pie(gender_count,  
  labels=names(gender_count),  
  main="Gender Distribution")
```

```
# Q3 Table
```

```
datatable(customer)
```

Q4 Dashboard Components

Bar Chart + Pie Chart + Table